Information Technology University, Lahore

Applied Machine Learning

BSCE23 Spring-2026

Assignment # 02

Issue Date: Friday 13/03/2026

Total marks =100

Instructions

1. **Please review the University Plagiarism Policy.**
2. **Any submissions handed in late will not be accepted and will receive a zero score.**
3. **This Assignment will access your CLOs as per OBE.**
4. **Assignment should be uploaded as PDF file with source file in Google Classroom.**
5. **The name of the file should be your Roll Number as; BSEEXXXX.**
6. **Please submit your own work only.**

Instructions for coding part:

- **Google Colab (Jupyter Notebook)** is recommended for the implementation of both models.

Problem Statement:

You are working as a data analyst in a healthcare research institute. The institute has collected diagnostic measurements from breast tumor samples and wants to build a system that can automatically determine whether a tumor is benign (non-cancerous) or malignant (cancerous).
Your task is to develop and compare machine learning classification models that can accurately predict whether a tumor is malignant or benign based on various medical measurements.
Using the Breast Cancer Wisconsin Dataset, implement and evaluate classification models to assist in early detection of breast cancer.

The Dataset

The Breast Cancer Wisconsin Diagnostic Dataset is publicly available and widely used in machine learning research.
Dataset Source:

https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

Dataset Characteristics:

- 569 samples
- 30 numerical features

Example features include:

- Mean radius
- Mean texture
- Mean perimeter
- Mean area
- Mean smoothness
- Mean compactness
- Mean concavity
- Mean symmetry

Target Variable:

- 0 → Malignant (Cancerous tumor)
- 1 → Benign (Non-cancerous tumor)

Your Task Requirements:

Implement and compare the following two classification methods:

- Naive Bayes
- Support Vector Machines (SVM)

Students must implement both approaches:

- From Scratch (Manual Implementation)
- Using Built-in Libraries (Scikit-learn)

Specific Tasks:

1. Data Preparation

- Load and explore the dataset
- Display basic dataset information
- Handle any necessary preprocessing
- Split the dataset into training and testing sets (70:30)

2. Model Implementation

Implement the following classifiers:

Naive Bayes

- Implement Gaussian Naive Bayes from scratch
- Then implement using sklearn GaussianNB

Support Vector Machine

- Implement SVM using scikit-learn
- Try both kernels:
- Linear Kernel
- RBF Kernel

3. Model Evaluation

For each model, calculate the following metrics:

- Accuracy Score
- Confusion Matrix
- Classification Report:
- Precision

- Recall
- F1-score

4. Comparative Analysis:

Answer the following questions:

1. Which algorithm achieved the highest accuracy?
2. How does SVM performance differ between linear and RBF kernels?
3. What are the strengths and weaknesses of Naive Bayes vs SVM?
4. Which model would you recommend for medical diagnosis systems and why?

Expected Deliverables:

- Make Report with results, comparison
- Complete Python code implementing in python file both classifiers.
- Performance comparison table showing metrics for both models
- Visualizations: Confusion matrices for both models
- Conclusion report with recommendations

Note: Feel free to ask for assistance if you encounter any issues.